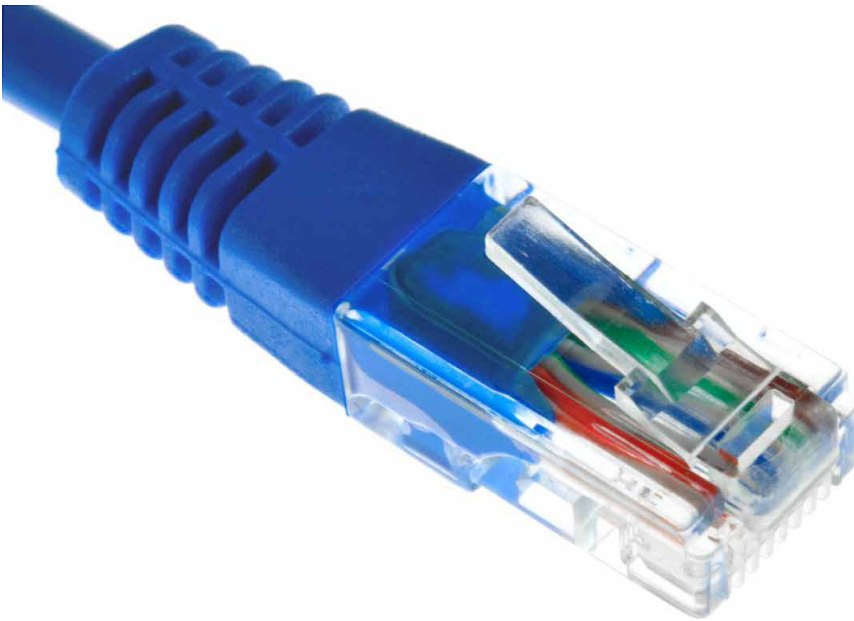


LabAmp network configuration guide



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Foreword

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1. Introduction

Products like the Kistler LabAmp amplifiers are very versatile devices and offer tremendous advantages by the fact that they provide an Ethernet interface for communication and can be accessed conveniently from any point within the network. This document gives a short overview about the different options how to use such devices conveniently in different situations/environments. A correct configuration is essential for an efficient and satisfying operation.

2. IP configuration options

2.1 For your safety

In a computer network, every device needs to have a unique address for a clear identification, the so called IP (internet protocol) address. There are two fundamental ways how the devices get their IP settings: static and dynamic.

Static IP configuration

- Each device within the network has a static IP address
- The IP settings need to be configured manually on each device in the network
- The user is responsible that no IP conflicts occur (every address only used once)

Dynamic IP configuration

- Each device in the network receives an IP address automatically
- The addresses are distributed by a DHCP server (dynamic host configuration protocol)
- If no DHCP server is present in the network, the IP addresses are automatically negotiated among the devices (Automatic Private IP Addressing (APIPA) or also AutoIP)

Depending on the situation, the different ways have different advantages and disadvantages

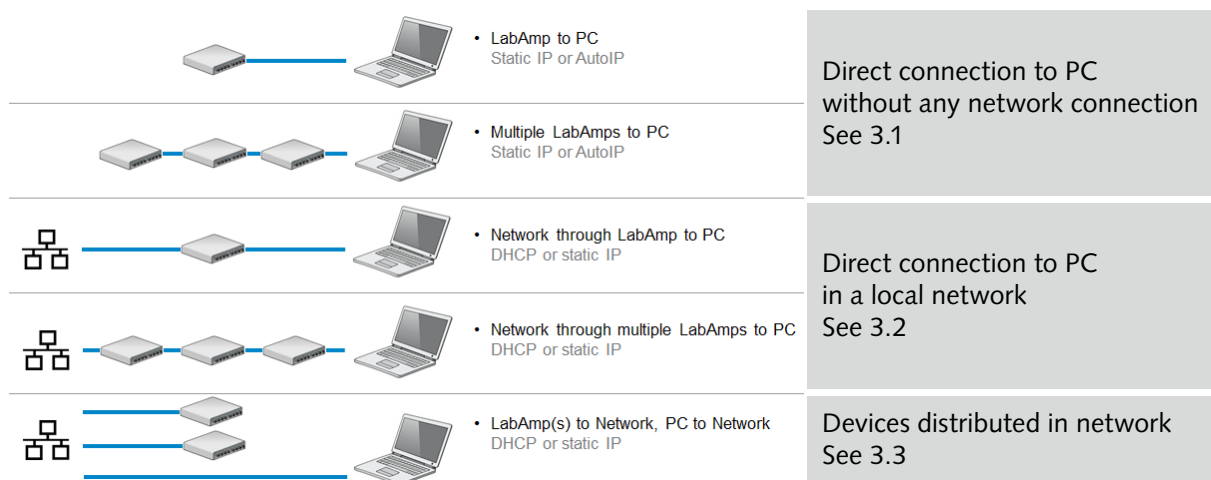


Fig. 1: Different network setups

3. Different setup options

3.1 Direct connection without network connection

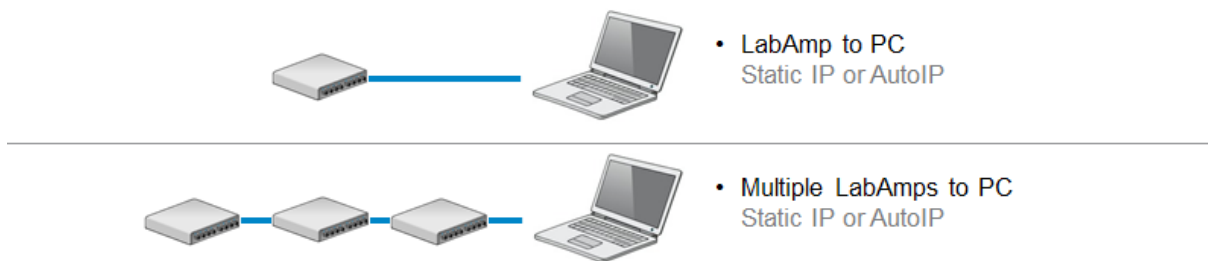


Fig. 2: Different examples of direct connections to a PC

Recommended situation

Static IP configuration

- If some devices are connected to a dedicated computer for certain measurements without the need to be connected to a company network or the internet at the same time.

AutoIP

- When a computer is used where the IP settings can't be changed to a static configuration due to missing permission.
- When a first connection needs to be established to one or more LabAmp devices.
- When just an ad-hoc measurement needs to be performed and the computer stays connected during this time.

Advantages

Static IP configuration

A static configuration is very convenient as every device has its known settings and keeps it even if power is switched off, cables are unplugged, etc. Addresses can be noted down and devices can be even labelled with their addresses so everyone knows how to connect quickly to each device

AutoIP

As the name already indicates, the mechanism works automatically, so no settings need to be configured manually. No administrator rights are required on the computer. After a certain time, the IP addresses are automatically determined and the devices are accessible. The actual addresses can be found by using the Kistler Network Setup Wizard.

Disadvantages

Static IP configuration

If the used computer is corporate IT equipment which is also needed for company network & internet access, it might be an inconvenient setup as not only the devices need to be in static IP configuration but also the used computer. As bigger company networks are normally using DHCP servers, the computer's IP settings would need to be changed every time it shall be connected to the LabAmps. Furthermore, restrictions by the IT administrators might even forbid changing the IP settings of a computer or accessing certain IP address ranges.

AutoIP

AutoIP, also known as APIPA (Automatic Private IP Addressing), is a fallback mechanism. Initially, a device in dynamic configuration waits for a DHCP server in the network to obtain an IP address. In a direct connection device(s)-to-computer, there is no such server. After a certain time with no answer, AutoIP is used to determine an IP address for each involved network device. This process can take up to 2 minutes which can be inconvenient.

Furthermore, when the devices are rebooted, the process starts from scratch and takes again some time. In addition, it is likely, that the created IP addresses have changed.

A short disconnection of the computer shouldn't lead to a loss of the IP addresses unless the computer's settings deactivate the Ethernet port and switch to Wifi as soon as there is no physical Ethernet connection anymore. In this case, the AutoIP process starts again once the network cable is connected again.

3.1.1 Options for direct connections

1. Every network interface has its own IP settings. In case the standard Ethernet interface of a computer is dedicated to access a company network, e.g. an external USB-to-Ethernet adapter can be used. This additional network interface can then be set to a static configuration to profit from above described advantages.



Fig. 3: A simple but useful USB-to-Ethernet adapter.
Picture source: Trendnet

2. As most notebooks nowadays offer network access via Wifi and Ethernet connection, the Ethernet interface can be set to a static configuration to profit from above mentioned advantages while the company network and internet can be still accessed using the Wifi network adapter.

However, sometimes the operating system or the computer's BIOS settings (e.g. HP EliteBooks, "LAN/WLAN switching" setting) block the use of Wifi and Ethernet at the same time. Normally, this can be configured. Ask your IT administrator to make the necessary changes if you don't feel comfortable to make changes in your computer's BIOS.

3.1.2 Example of a static configuration

A valid configuration for a static IP setting is the following:

Computer settings:

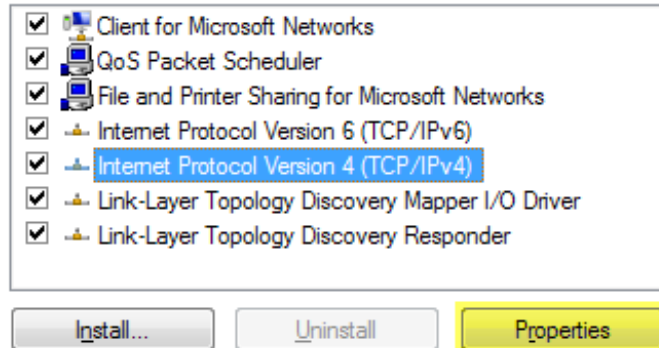
IP-address:	192.168.1.100
Subnet mask:	255.255.255.0
Default gateway:	192.168.1.1

LabAmp settings:

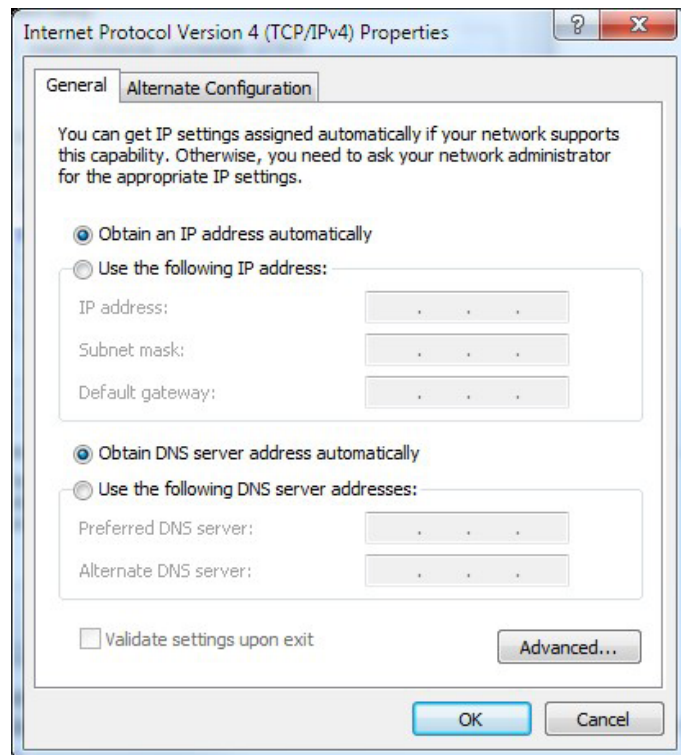
IP-address:	192.168.1.101 (for further devices count up the last number; 102, 103, etc.)
Subnet mask:	255.255.255.0
Default gateway:	192.168.1.1

The LabAmp settings can be configured either by using the Kistler Network Setup Wizard or on the individual web user interface of every device. The computer settings need to be configured in the control panel → Network And Sharing Center → LAN connection → Properties:

This connection uses the following items:



→ select Properties:



Here the IP settings of the computer can be modified.
ock diagram

3.2 Direct connection without network connection

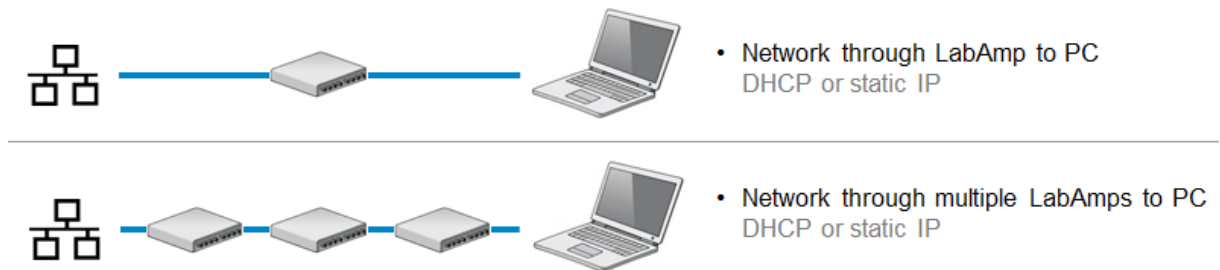


Fig. 4: LabAmps in a local network and PC in daisy-chain

Normally the user doesn't have an influence on the architecture of the local network e.g. a company network. Therefore we don't compare the options of different network setups but focus on the most common case when there is a DHCP server present in the network.

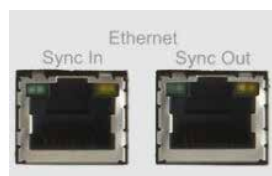
Anyway, in case a local network is using static IP settings, the configuration is exactly the same as described under 3.1, "Static IP Configuration".

Remark: Do not connect devices with static IP configuration to networks with dynamic IP configuration. This can lead to IP address conflicts

Recommended situation

DHCP

If measurements and configurations shall be made on one or more normal computers in a company network, connecting the LabAmp devices to a company network where IP addresses are managed by a DHCP server, this is the best option. The two Ethernet interfaces with integrated switch functionality allow wiring the company network through the LabAmp devices so that the actual PC can be connected to the last device in the daisy chain.



Note: If a LabAmp device is switched off, the integrated switch is turned off, too.

Advantages

DHCP

After connecting the devices, they automatically obtain an IP address from the DHCP server in the network. The Kistler Network Setup Wizard shows the addresses quickly and work can start without any further IP configurations.

Furthermore, the devices are accessible within the whole company network so once the IP addresses are known, working with them is possible from any point in the network. In a big company this can involve different countries and continents. Also see 3.3.

It is basically possible to allocate the same IP address to a device each time it is connected to the network based on their unique MAC address. However, these are advanced DHCP server settings which need to be configured by your network administrator.

Disadvantages

DHCP

- Some companies have restrictions regarding the connection of "unapproved" devices to the network. In such cases, a discussion with the IT department might help. LabAmp devices do not act as DHCP server and they do not contain any Wifi access point functionality which could disturb the network. If they are set to DHCP mode, they do not interfere with the network management.
- LabAmp devices are laboratory equipment designed with ease of use and quick operation in mind. This is why there is no authorization mechanism in place. The person who has access can work with the device without any restriction. If LabAmp devices are used in very sensitive areas where the possibility of unauthorized access has to be avoided, this must be solved by the network administrator.

3.3 DHCP Configuration, PC somewhere in the network



Fig. 5: LabAmps somewhere in the network

Remark: Do not connect devices with static IP configuration to networks with dynamic IP configuration. This can lead to IP address conflicts.

Recommended situation

DHCP

Wherever LabAmp devices need to be accessed from a place within the network which is physically further away, this is the perfect setup. This can be the case if a measurement shall be monitored out of a different room or building.

Advantages

DHCP

There are basically no distance restrictions. A device can be accessed from another part of the world if it is located in the same (company) network. The devices obtain their IP settings from the DHCP server within the network.

It is basically possible to allocate the same IP address to a device each time it is connected to the network based on their unique MAC address. However, these are advanced DHCP server settings which need to be configured by your network administrator.

Disadvantages

DHCP

If the LabAmp devices are placed in a different subnet compared to the computer from where the devices shall be accessed, the Kistler Network Setup Wizard isn't able to find the devices. In this case, either a computer has to be connected to the same subnet running the Network Setup Wizard to find the device(s) or the network administrator has to inform about the distributed IP addresses.

Other than that, same advantages/disadvantages listed under 3.2 apply.

4. First connection via ethernet

The measuring amplifiers Type 5167A and 5165A include an integrated web server, whose web interface is used to configure and operate the device. Please ensure that your PC is in DHCP mode and/or Auto-IP mode (network settings) when connecting to a 5167A or a 5165A for the very first time.

The following approach is recommended to establish a connection with the measuring amplifier via Ethernet:

1. Connect the device to the network

Connect the measuring amplifier via a network cable directly to a computer or a network (e.g network switch).

2. Find the device in the network

- Start the «IP Setup Wizard» tool that finds all Kistler network devices (the tool can be found on www.kistler.com, search for «5167» or «5165» or click here).



Fig. 6: Start screen of the Kistler Network Setup Wizard

- Click on «Network Setup Wizard» and wait until Kistler network devices are displayed. In the case of a direct connection (5165A or 5167A to PC), it may take up to 2 minutes until the devices are displayed.
- The table displays all, previously not configured Kistler network devices.

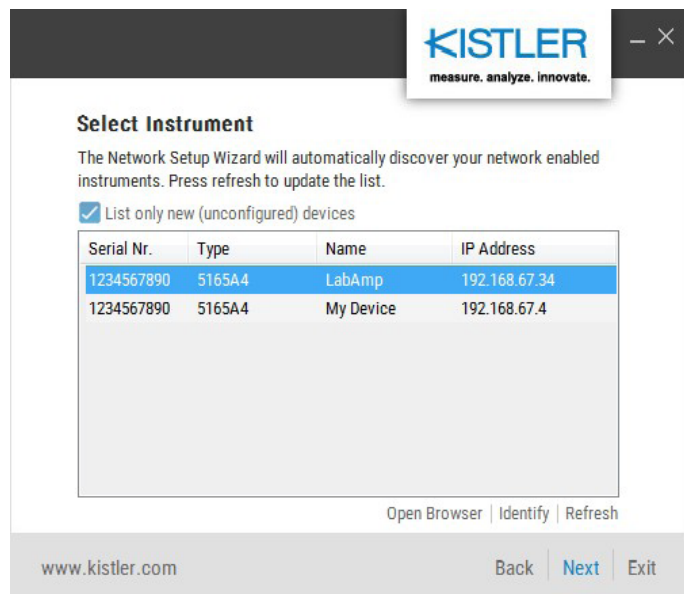


Fig. 7: Network Setup Wizard, list with found devices

To see all devices, including the already configured ones, deselect the «List only new (unconfigured) devices» checkbox. It might take up to 2 minutes to have your device appear in the list after connecting to the network.

3. Access to the device via web browser

Bottom right of the list field you see different options:

- «Open browser»
Opens the web interface of the respective device in your standard web browser. The Kistler LabAmp 5167A can now be configured and operated via the web interface.
- «Identify»
Permits an identification of the selected device. The identified measuring amplifier signals this through the yellow flashing of all LEDs for approx. 4 s. This function is very useful if several measuring amplifiers are used in the same network.
- «Refresh»
Repeats the search and refreshes the list of devices found in the network.

Adapting network settings

If required, the network settings for the marked device (see Figure 7: Network Setup Wizard, list with found devices) can be adapted by clicking on «Next». The following options can be configured for this purpose:

- **Device name:**
The device name is displayed in the Network Setup Wizard as well as in the web interface. In networks that support name resolution, the device name can be entered directly in the address field of the web browser to access the device.
- **Address mode**
The address modes «DHCP with Auto-IP» and «Static IP» are supported. Configure the network settings as desired and click on «Next» until you reach the final review page. To apply the changes, press «Update»

Information regarding network connection

The Kistler Network Setup Wizard finds Kistler network devices throughout the network. It even scans multiple network interfaces on a computer. However, if there are several routers or switches between the computer and the measuring amplifier, this search might not be successful under certain circumstances.